Study Guide: Scale and Architecture in Distributed Systems

This study guide provides a comprehensive review of the fundamental concepts, scalability challenges, and architectural styles associated with distributed systems as outlined in the provided lecture materials.

Part 1: Short-Answer Quiz

Instructions: Answer the following questions using 2-3 sentences based on the information provided in the source text.

1. **What are the three primary components used to measure scalability in a distributed system?**
2. **Why is transitioning a distributed system from a Local Area Network (LAN) to a Wide Area Network (WAN) technically difficult?**
3. **What is the "essence" of administrative scalability challenges?**
4. **How do peer-to-peer networks like BitTorrent or Skype manage to bypass some administrative scalability issues?**
5. **What are the three root causes of scalability problems in centralized solutions?**
6. **Explain the technique of hiding communication latencies to improve size scalability.**
7. **What is the primary problem that arises when implementing replication or caching in a system?**
8. **How does global synchronization affect the development of large-scale distributed solutions?**
9. **What characterizes the Publish/Subscribe architectural style?**
10. **What are the specific characteristics that distinguish the Shared Dataspace architectural style from other models?**

Part 2: Answer Key

1. **Scalability Components:** Scalability is comprised of size scalability (number of users and processes), geographical scalability (maximum distance between nodes), and administrative scalability (number of administrative domains). Most modern systems focus primarily on size scalability, though geographical and administrative factors present the greatest current challenges.
2. **LAN to WAN Transition:** WAN links are often inherently unreliable and suffer from latencies that prohibit synchronous client-server schemes where a client waits for a response. Additionally, WANs lack multicast communication, meaning simple broadcasts cannot be deployed without separate directory services.
3. **Administrative Scalability Essence:** The essence of this challenge lies in conflicting policies between different domains. These conflicts typically concern usage and payment, management strategies, and security protocols.
4. **P2P Network Collaboration:** In networks such as BitTorrent, Skype, and Spotify, the end users collaborate directly with one another rather than through administrative entities. This allow these systems to function as exceptions to traditional administrative scaling limitations.
5. **Centralized Scalability Root Causes:** Scalability in centralized systems is limited by the computational capacity of the CPUs and the storage capacity, including the transfer

rate between CPUs and disks. Furthermore, the network capacity between the user and the centralized service acts as a significant bottleneck.

6. **Hiding Latencies:** This technique involves avoiding waiting for responses by making use of asynchronous communication. A system can have a separate handler for incoming responses, allowing it to perform other tasks while the communication is processed.
7. **Consistency and Inconsistency:** Applying replication or caching leads to inconsistencies because modifying one copy of data makes it different from the others. While inconsistencies can sometimes be tolerated depending on the application, they often require complex management to keep data synchronized.
8. **Global Synchronization Impact:** Keeping all copies of data consistent generally requires global synchronization on every modification. This requirement is a significant barrier because it precludes the implementation of truly large-scale solutions.
9. **Publish/Subscribe Style:** This architectural style is based on the decoupling of processes, which makes the interactions "anonymous." It utilizes an event bus where components publish information and other components receive event deliveries.
10. **Shared Dataspace Style:** The shared dataspace style is characterized by being both anonymous and asynchronous. It utilizes a shared, persistent data space where components can publish and deliver data without direct time constraints or process identification.

Part 3: Essay Questions

Instructions: Use the concepts from the source text to develop detailed responses to the following prompts.

1. **The Scalability Triad:** Analyze the three components of scalability—size, geography, and administration. Discuss why developers often prioritize size scalability and the specific technical hurdles that make geographical and administrative scalability more difficult to achieve.
2. **Trade-offs in Replication:** Evaluate the benefits of replication and caching against the challenges of data consistency. In your discussion, explain why global synchronization is often incompatible with the goals of a large-scale distributed system.
3. **Communication Models and Scale:** Compare synchronous and asynchronous communication. Explain how the choice of communication model directly impacts a system's ability to scale across a Wide Area Network (WAN).
4. **Evolution of Architectural Styles:** Describe how the need to decouple processes and remove time constraints led to the development of "anonymous" and "asynchronous" architectural styles. Contrast these with traditional Layered and Object-based styles.
5. **Distributed Resources and Policy:** Examine the role of administrative scalability in the context of computational grids and shared equipment. Discuss the difficulties of managing shared resources when multiple administrative domains have conflicting usage and security policies.

Part 4: Glossary of Key Terms

- **Administrative Scalability:** The ability of a distributed system to span multiple independent administrative domains with conflicting policies.
- **Asynchronous Communication:** A communication method where the sender does not wait for a response before continuing with other tasks, often utilizing separate response handlers.
- **Caching:** The process of making copies of data available at or near the client (e.g., in browsers or proxies) to reduce access time and server load.
- **Computational Grid:** A system that shares expensive or high-performance resources across different administrative domains.
- **Geographical Scalability:** The ability of a system to function effectively despite the physical distance between its nodes, overcoming issues like latency and unreliable WAN links.
- **Global Synchronization:** The process of ensuring all copies of data in a distributed system are modified simultaneously to maintain consistency; this often limits total system scale.
- **Layered Style:** An architectural style commonly used for client-server systems where components are organized into logical layers.
- **Object-based Style:** An architectural style used for distributed object systems where components interact through method calls.
- **Publish/Subscribe:** An architectural style where processes are decoupled and anonymous, communicating via an event bus.
- **Replication:** The technique of making copies of data available on different machines, such as mirrored websites or replicated databases, to improve availability and scale.
- **Shared Dataspace:** An architectural style that is both anonymous and asynchronous, relying on a shared, persistent space for data exchange.
- **Size Scalability:** The ability of a system to handle an increasing number of users and/or processes without a degradation in performance.
- **Synchronous Interaction:** A communication model where a client sends a request and must wait for an answer from the server before proceeding.